Aaron Hayward Dost

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Professional Summary:

Full Stack Developer specializing in Next.js, TypeScript, and CMS integrations. Proven track record of modernizing enterprise websites, improving performance, and aligning technical solutions with business objectives. Passionate about building scalable and maintainable web applications.

Technical Skills:

- **Languages:** TypeScript, JavaScript (ES6+), HTML, CSS, Rust, SQL, GLSL
- **Frameworks/Libraries:** React, Preact, Next.js, Tailwind CSS, Node.js, NestJS, WebGL2
- **Backend/Tools:** Payload CMS, NetSuite, Google Tag Manager, Git, Tauri, Docker, Mongo DB, Redis Streams, N-API

Education:

Conestoga College - Software Engineering Technology - April 2024
Conestoga College - Business Administration: Finance - April 2019

Experience:

Full Stack Developer | Easy Kleen Pressure Systems Ltd. | April 2024 - Current

- Migrated legacy WordPress site to a modern Next.js/Payload CMS stack, engineering a responsive Tailwind frontend that drove a 50% increase in mobile engagement.
- Optimized performance for high-traffic product pages, reducing load times from 1.1s to 200ms through intelligent caching strategies (ISR) and optimized database queries.
- Architecting a multi-tenant CMS infrastructure to centralize content management across the corporate umbrella, enabling a single codebase to drive distinct brand experiences and significantly reduce maintenance overhead.
- Engineered a secure B2B Dealer Portal deeply integrated with the NetSuite REST API, providing authenticated resellers real-time access to invoices, inventory data, and customer metrics.
- Digitized sales operations by developing a self-service quote submission system, automating ERP data entry and eliminating the dependency on manual phone-in orders for standard requests.

Founder | Totally Legit Media Group | Social Media Management | February 2020 - December 2023

- Founded and scaled a service-based agency to $100K annual revenue while completing full-time studies.
- Led a cross-functional team of 8, managing project timelines, client deliverables, and resource allocation.

Projects:

Kerr Black Hole Simulation Engine | Advanced Physics Simulation | [Source code walkthrough upon request]

- Created a hybrid rendering pipeline decoupling high-precision physics (Rust/f64) from real-time presentation (WebGL2), solving floating-point precision limitations inherent to browser-based rendering.
- Engineered a custom numerical integrator using 8th-order Runge-Kutta (DOPRI8) with adaptive stepping to solve null geodesics in curved spacetime, achieving sub-1e-12 error tolerances.
- Designed a build-time asset generation CLI that parallelizes ray tracing across CPU cores (Rayon), compressing gigabytes of simulation data into optimized binary transfer maps (RGBA32F) for low-latency client streaming.
- Implemented robust statistical validation using Mahalanobis distance and covariance matrix analysis to automatically detect and classify integration anomalies in 5D phase space.

MMO Game Server | NestJS Multiplayer Game Server | [Source code walkthrough upon request]

- Designed a distributed, event-driven backend to support 25k+ concurrent users (CCU), utilizing Redis Streams to implement a custom Event Sourcing pattern (Global Fact Bus) that ensures deterministic state across regional shards.
- Devised a hybrid runtime strategy to overcome Node.js Garbage Collection limits, formulating a high-performance architecture that offloads CPU-intensive mathematics (AOI, collision) to Rust via N-API while retaining Nest.js for business logic orchestration.
- Outlined a CQRS-based communication protocol, decoupling client intent (Commands) from global state (Facts); replaced Socket.io with uWebSockets.js to minimize I/O overhead and achieve sub-10ms latency targets.
- Engineered a strict Monorepo architecture managing 50+ complex Mongoose schemas, enforcing end-to-end type safety across the Gateway, Orchestrator, and Gameplay services using shared TypeScript generics.